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Subject: Programming with Java	
ASSIGNMENT No : 1	
TITLE : Java Environment Setup and Basic Program Execution	

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

Programme

Output

Conclusion

This assignment helped us get a solid understanding of the Java ecosystem through the study of JDK, JRE, and JVM, along with the practical implementation of variables, methods, and constructors using suitable examples. The concepts demonstrated how Java follows an object-oriented approach where constructors initialize objects, methods define behaviour, and variables maintain program state. Overall, this helped us establish a strong foundation in core Java programming that serves as the basis for building real-world applications.

Questions

- 1. Explain the main concept used in Java environment setup, variables and methods.**

Java environment setup involves installing the Java Development Kit (JDK) and configuring the PATH and JAVA_HOME environment variables so the system can locate the Java compiler (javac) and runtime (java). Variables in Java are typed memory locations that store data values and must be declared with a specific data type such as int, float, or String before use. Methods are named blocks of code that

perform specific tasks and can accept parameters and return values, promoting reusability and modularity. Together, these form the fundamental building blocks of any Java program, allowing structured and object-oriented development.

2. What are the advantages of using Java variables and methods?

Java variables provide strongly typed data storage, which reduces runtime errors by catching type mismatches at compile time itself, making programs more reliable and predictable. Methods promote code reusability by allowing the same logic to be called multiple times from different parts of the program without rewriting it, reducing redundancy. The use of local and global variables along with method parameters enables better memory management and controlled access to data across the program. Overall, variables and methods together make Java programs more organized, readable, maintainable, and easier to debug.

3. How can this implementation be improved?

The implementation can be improved by incorporating exception handling using try-catch blocks to gracefully manage runtime errors such as invalid input or out-of-bounds access instead of letting the program crash. Input validation can be added to ensure that variables receive only appropriate values before being processed by methods, improving program robustness. Methods can be further modularized by following the single responsibility principle, where each method performs exactly one specific task, making the code cleaner and easier to test. Additionally, using access modifiers such as private and public appropriately would improve encapsulation and follow object-oriented best practices.

4. What are the real-world applications?

Java variables and methods form the foundation of virtually every real-world application software, from Android mobile app development to large-scale enterprise backend systems. In banking and financial software, typed variables ensure precise handling of monetary values while methods encapsulate business logic such as interest calculation and transaction processing. E-commerce platforms use Java methods extensively for cart management, payment processing, and user authentication workflows. Java is also widely used in embedded systems, scientific computing, and web development frameworks like Spring, all of which rely heavily on well-structured variables and methods for efficient and scalable functionality.